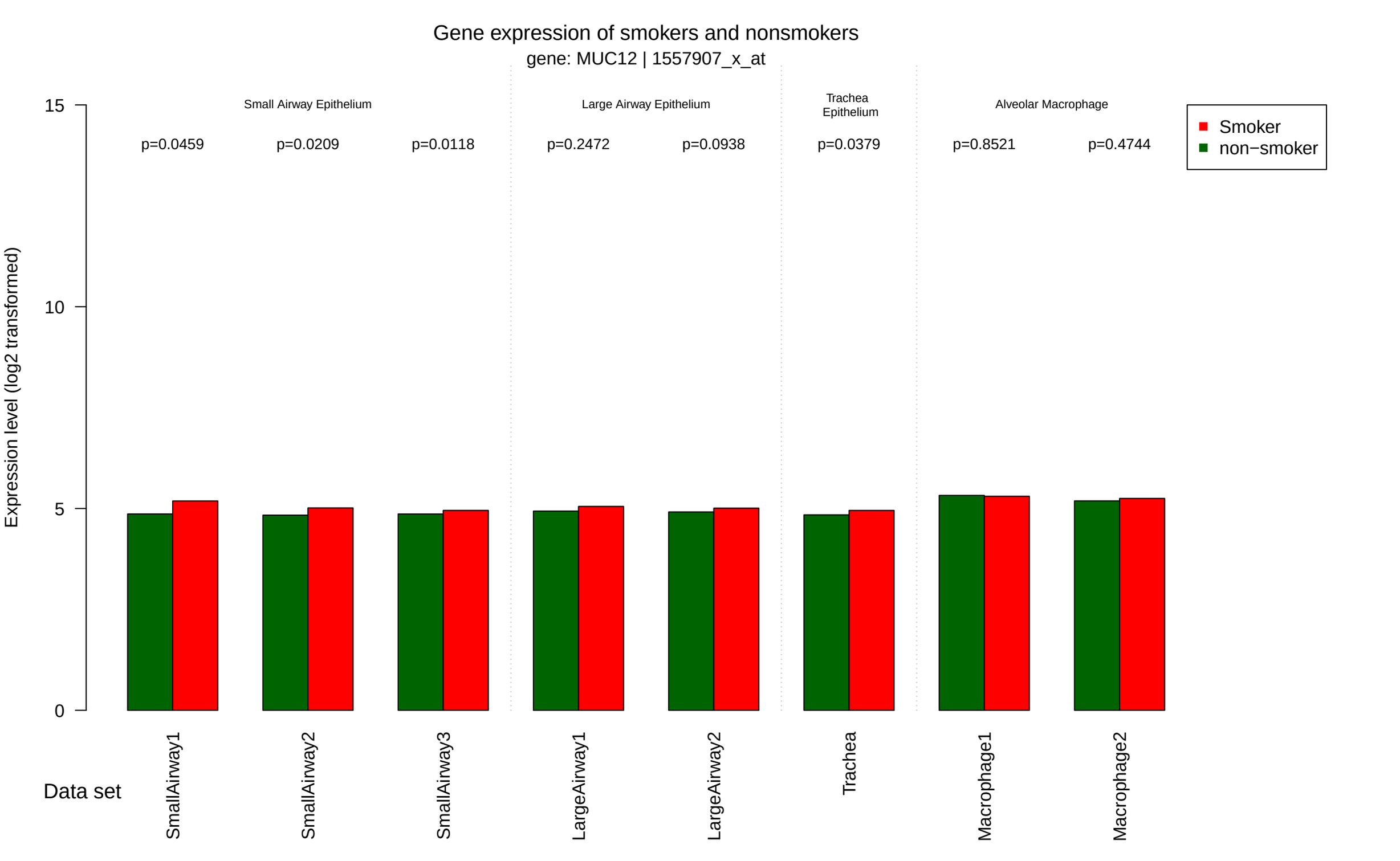
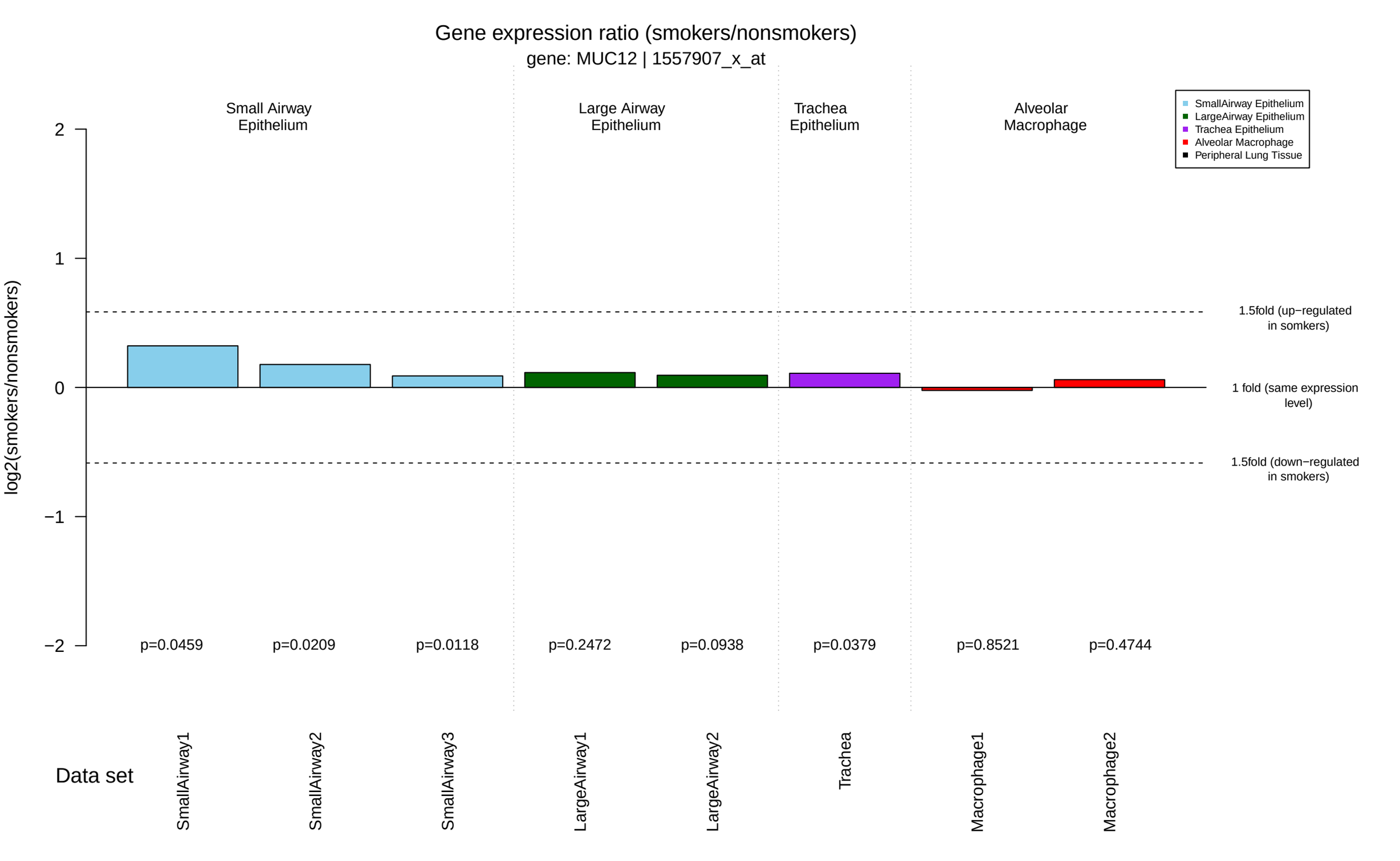
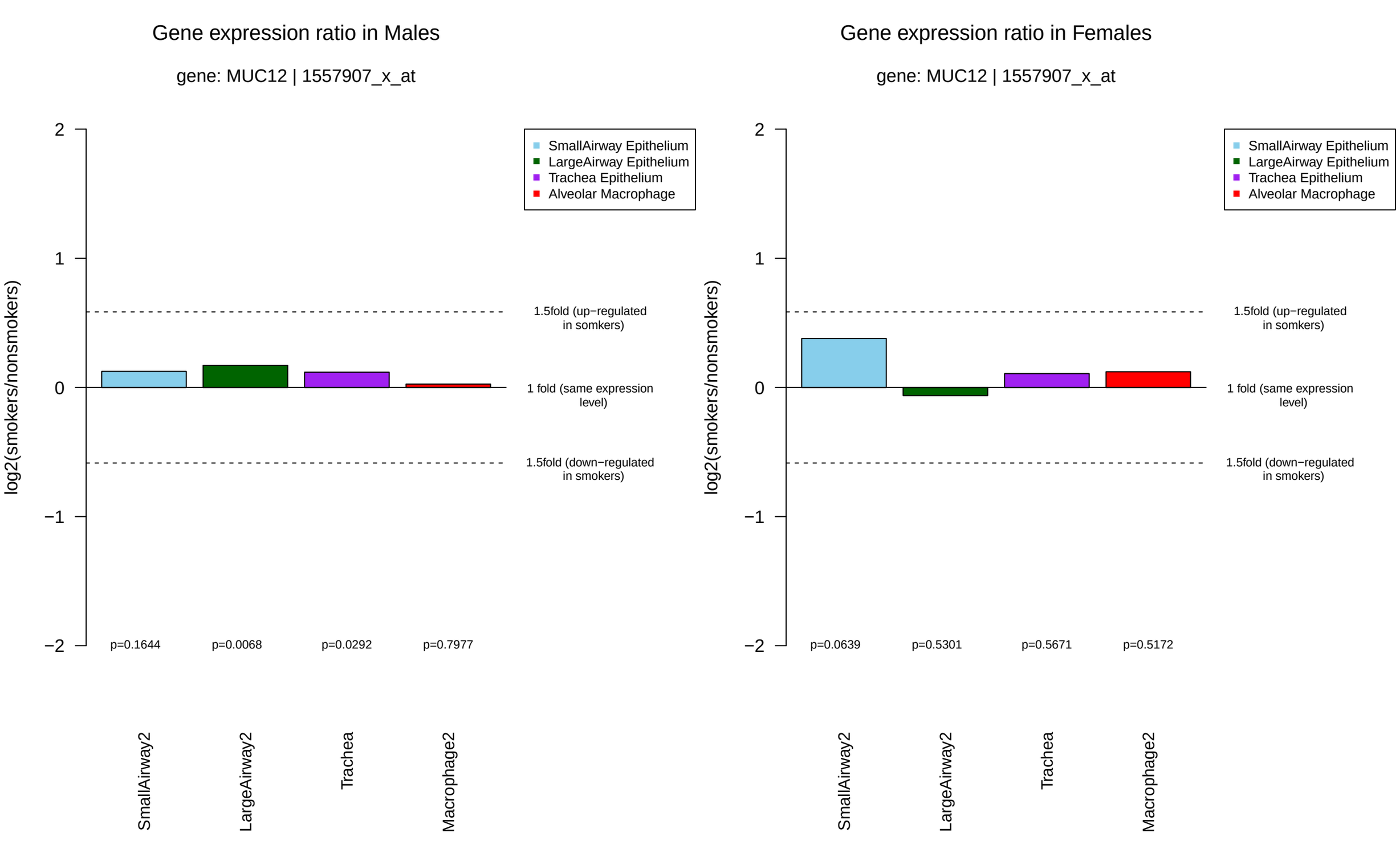
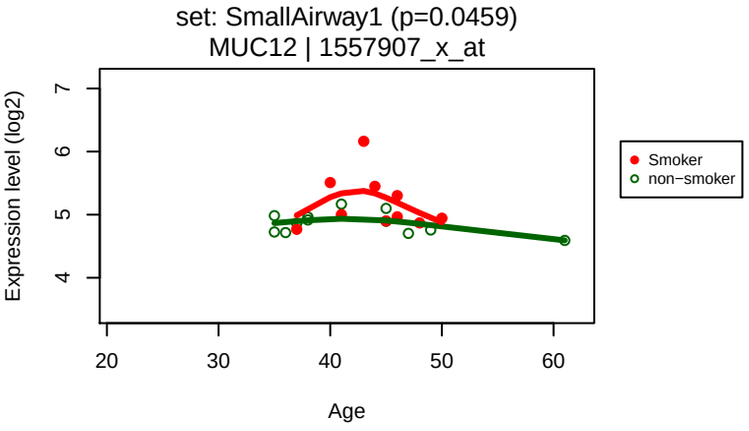
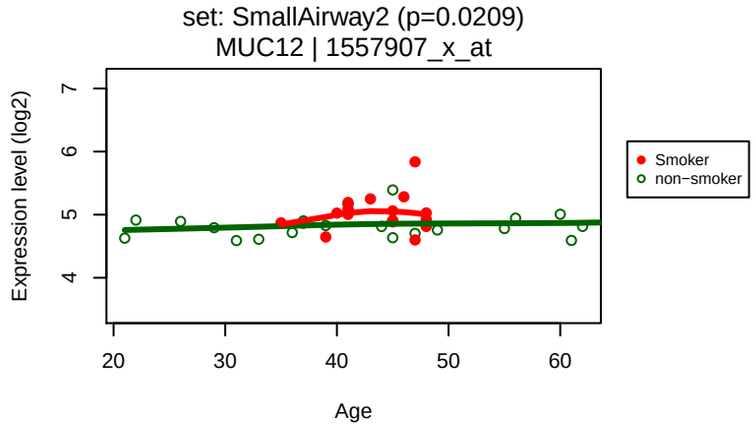
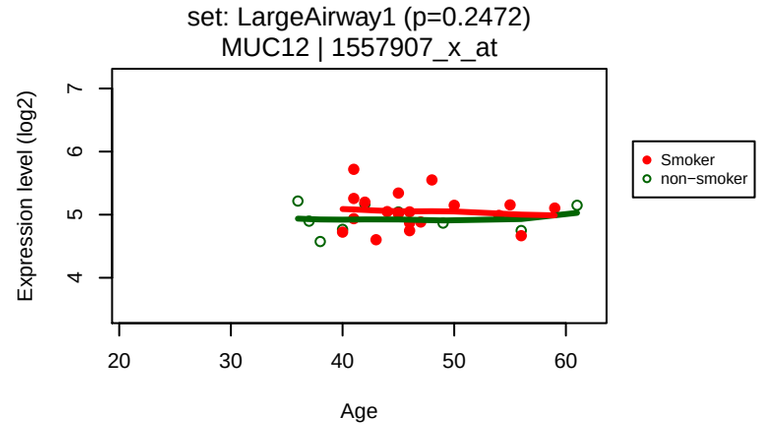

Gene expression of smokers and nonsmokers
gene: MUC12 | 1557907_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0459
p=0.0209
p=0.0118
p=0.2472
p=0.0938
p=0.0379
p=0.8521
p=0.4744
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

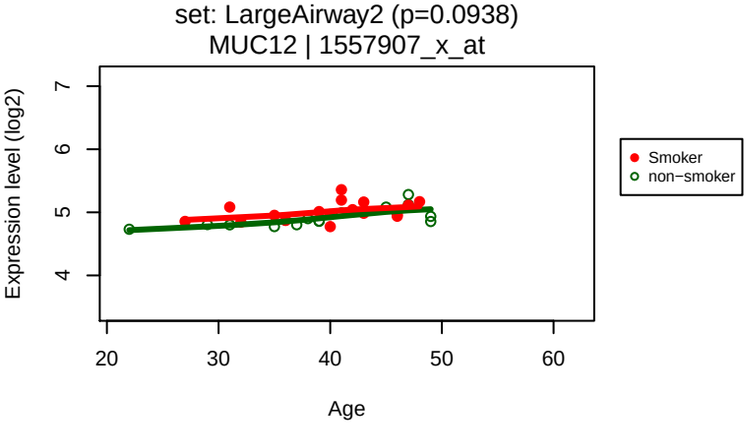

Gene expression ratio (smokers/nonsmokers)
gene: MUC12 | 1557907_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0459
p=0.0209
p=0.0118
p=0.2472
p=0.0938
p=0.0379
p=0.8521
p=0.4744
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

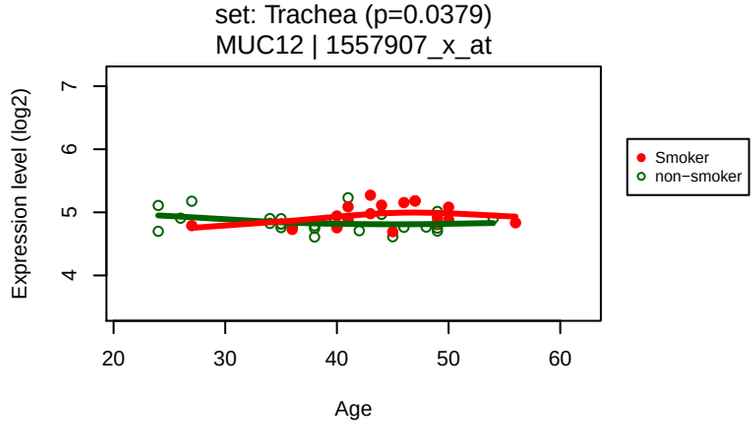

Gene expression ratio in Males
gene: MUC12 | 1557907_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1644
p=0.0068
p=0.0292
p=0.7977
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC12 | 1557907_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0639
p=0.5301
p=0.5671
p=0.5172
SmallAirway2
LargeAirway2
Trachea
Macrophage2

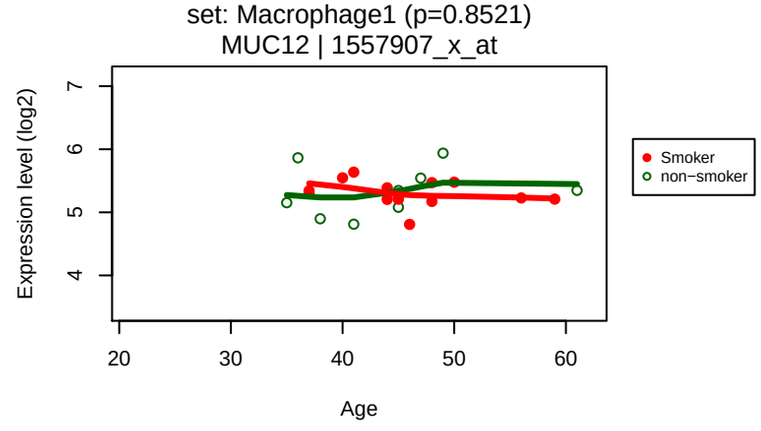

set: SmallAirway1 (p=0.0459)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker

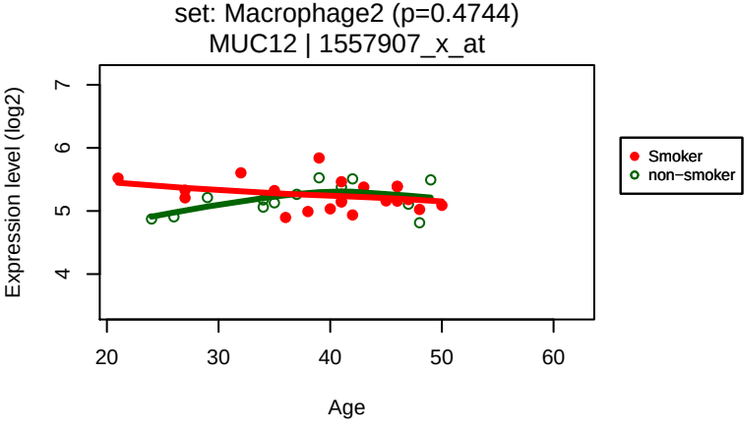

set: SmallAirway2 (p=0.0209)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker

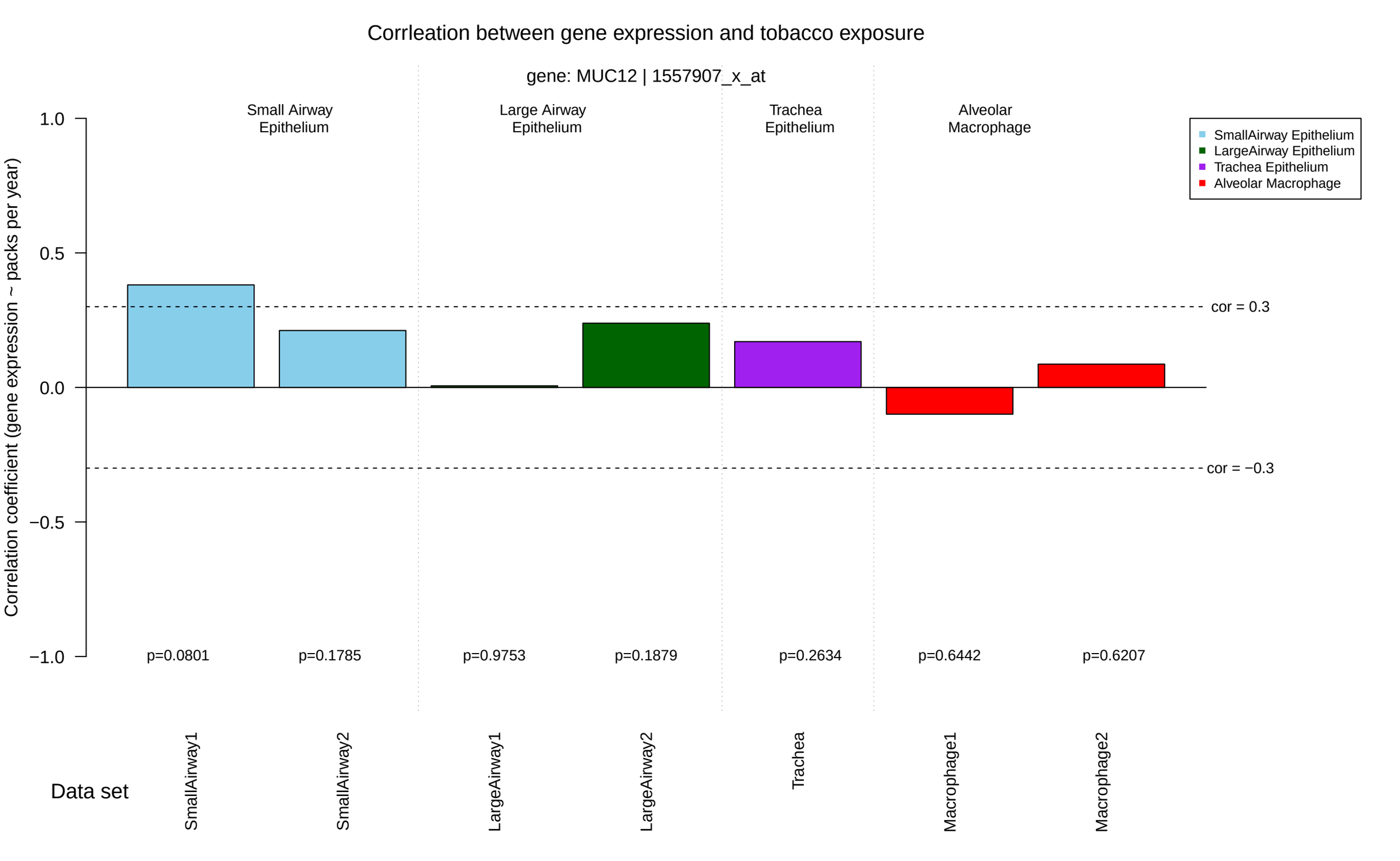

set: LargeAirway1 (p=0.2472)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0938)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0379)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8521)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4744)
MUC12 | 1557907_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC12 | 1557907_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0801
p=0.1785
p=0.9753
p=0.1879
p=0.2634
p=0.6442
p=0.6207
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2